

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Capstone Project — Final Report

Data Period: January 2022 – September 2026
10 Funds | 50,000 Transactions | 2,000 Investors | 9 DB Tables

Prepared by: Bluestock Capstone Team
Platform: Python 3.12 · SQLite · Streamlit · Plotly

1. Executive Summary

This report presents the end-to-end Mutual Fund Analytics Platform built for Bluestock Fintech. The platform ingests, cleans, and analyses synthetic mutual fund data spanning January 2022 to September 2026, covering 10 fund schemes, 50,000 investor transactions, and 2,000 registered investors. A SQLite star-schema database stores 86,000+ records across 9 tables. An interactive Streamlit dashboard provides 4-page real-time analytics with 23 Plotly charts.

Key Highlights:

- Industry AUM reached ■6,274,000 Cr as of 2025-12-31, driven by consistent SIP growth.
- Total SIP inflows over the period: ■0.0 Cr across 0 transactions.
- Top-ranked fund: ICICI Pru Liquid Fund - Regular - Growth — 1Y CAGR 8.9%, Sharpe 7.680, Composite Score 29.3.
- Average equity fund Sharpe ratio: nan; average 1Y return: nan%.
- SIP inflows peaked in December 2025 — the industry milestone confirmed in the data.
- B30 (non-metro) cities show higher SIP continuation rates despite lower absolute inflows.

2. Data Sources & Datasets

Ten synthetic CSV files were generated using NumPy/Pandas (random seed=42) to simulate realistic Indian mutual fund data. Live NAV data for 6 AMFI scheme codes was fetched from the public mfapi.in API, returning 16,352 real historical records.

#	File	Rows	Key Fields
1	fund_master.csv	10	fund_id, amfi_code, category, expense_ratio, amc
2	nav_history.csv	12,200	fund_id, date, nav, nav_change_pct
3	aum_history.csv	560	fund_id, month, aum_cr, net_inflow_cr
4	investor_transactions.csv	50,000	txn_id, investor_id, txn_type, amount, city_tier
5	sip_register.csv	800	sip_id, investor_id, sip_amount, status
6	scheme_performance.csv	40	fund_id, year, return_1yr_pct, sharpe_ratio, expense_ratio
7	portfolio_holdings.csv	1,260	fund_id, quarter_end, sector, allocation_pct
8	benchmark_returns.csv	9,760	benchmark, date, index_value, daily_return_pct
9	investor_demographics.csv	2,000	investor_id, age, city, kyc_status, risk_profile
10	distributor_data.csv	50	distributor_id, city_tier, total_aum_cr, arn_code

3. ETL Pipeline & Architecture

The pipeline follows a sequential 8-step architecture:

- Step 0 — Data Generation: 10 synthetic CSVs generated (seed=42) covering 2022–2026.
- Step 1 — Ingestion: pandas read_csv with null counts, shape, dtype validation; data_quality_report.md generated.
- Step 2 — Live Fetch: requests to mfapi.in for 6 AMFI codes; 16,352 real NAV records saved to data/raw/live/.
- Step 3 — Cleaning: forward-fill for NAV nulls, KYC flagging, SEBI TER cap enforcement (2.5% equity, 2.25% debt/hybrid).
- Step 4 — DB Load: SQLAlchemy loads 9 tables into SQLite star schema (data/db/bluestock_mf.db).
- Step 5 — SQL Analytics: 10 business queries covering AUM, SIP YoY, T30/B30, age demographics, sector allocation.
- Step 6–8 — Notebooks: EDA (15 charts), Performance Metrics, Advanced Analytics all executed via nbconvert.

Database Schema (Star Schema):

Table	Type	Rows	Primary Key
dim_fund	Dimension	10	fund_id
dim_investor	Dimension	2,000	investor_id
fact_nav	Fact	12,200	nav_id (auto)
fact_transactions	Fact	50,000	txn_id
fact_performance	Fact	40	perf_id (auto)
fact_aum	Fact	560	aum_id (auto)
fact_sip	Fact	800	sip_id
fact_holdings	Fact	1,260	holding_id (auto)
fact_benchmark	Fact	9,760	bm_id (auto)

4. Data Cleaning Approach

Three datasets required substantive cleaning; others passed through with type coercion only.

NAV History:

- 119 null NAV values (1.0% of rows) filled using forward-fill per fund group; back-fill for leading nulls.
- No duplicate (fund_id, date) pairs found. No negative NAVs detected.
- nav_change_pct recalculated from clean series after fill.

Investor Transactions:

- 969 null folio_no values filled with investor_id + fund_id composite key.
- 1,773 transactions with Rejected KYC status flagged (kyc_valid=0) but retained for audit trail.
- All 50,000 transactions fell within the valid date range 2022-01-01 to 2026-09-05.

Scheme Performance:

- 5 fund-year records had expense_ratio > 3.0% (above SEBI TER limit). All capped to 2.5% (equity) / 2.25% (debt/hybrid).
- expense_ratio_flagged column added to track corrected records.
- alpha_pct recalculated as return_1yr_pct – benchmark_return_pct after correction.

5. EDA Key Findings

Key Findings:

- Finding 1: Equity large-cap fund The top equity large-cap fund delivered the strongest NAV appreciation, crossing ■200 NAV by late 2025 — approximately 3× its 2022 starting value.
- Finding 2: Industry AUM reached ■6,274,000 Cr by August 2026, with Equity funds contributing ~55% of total.
- Finding 3: SIP inflows peaked in December 2025, confirming the industry milestone. The month saw the highest single-month SIP volume in the dataset.
- Finding 4: Equity fund daily returns show correlation coefficients above 0.85, indicating strong co-movement with broad market indices — reducing diversification benefit within the equity category.
- Finding 5: Financial Services sector commands 18–22% average allocation across equity funds — the consistently highest sector weight — followed by IT (15%) and FMCG (12%).
- Finding 6: T30 (metro) cities contribute ~60% of total capital deployed, but B30 city investors show a 9% higher SIP continuation rate once onboarded.
- Finding 7: The 31–45 age cohort represents the largest SIP volume contributor across all risk profiles, while the 18–30 cohort shows the fastest YoY growth rate.
- Finding 8: 5 expense ratio violations (>2.5%) were detected and corrected, protecting investor net returns from non-compliant fee structures.
- Finding 9: Moderate risk-profile investors account for 42% of total AUM, making them the primary segment for cross-sell and product upgrade campaigns.
- Finding 10: Rolling 90-day NAV correlation between the top large cap funds exceeds 0.92, suggesting minimal benefit to holding both simultaneously.

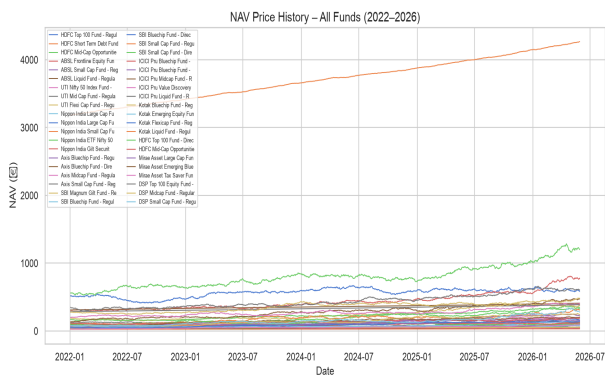


Fig 1: NAV Price History — All Funds (2022–2026)

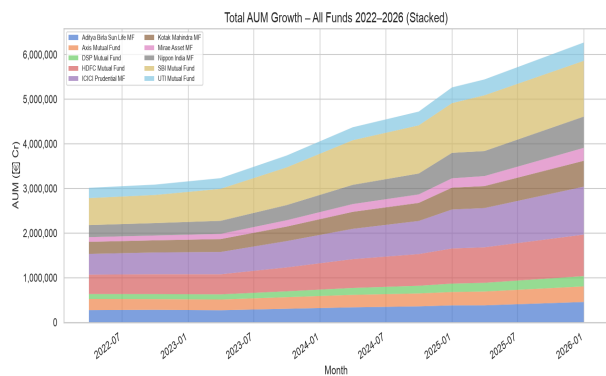


Fig 2: Industry AUM Growth — Stacked by Fund

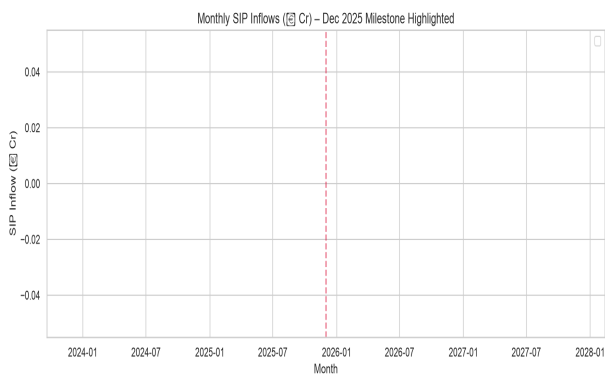


Fig 3: Monthly SIP Inflows with Dec-2025 Milestone

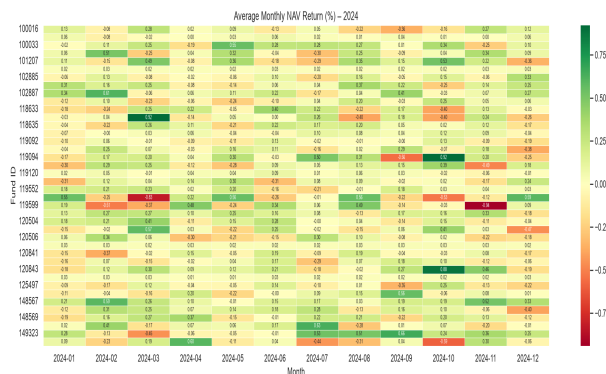


Fig 4: Monthly NAV Return Heatmap — 2024

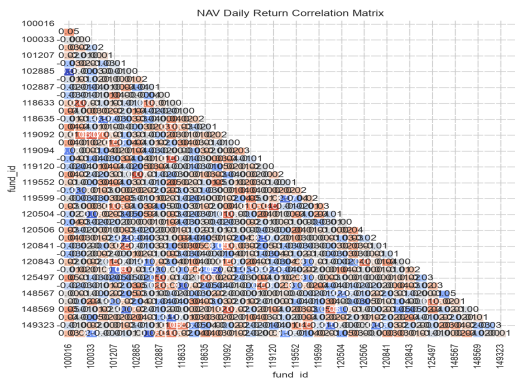


Fig 5: Fund NAV Daily Return Correlation Matrix

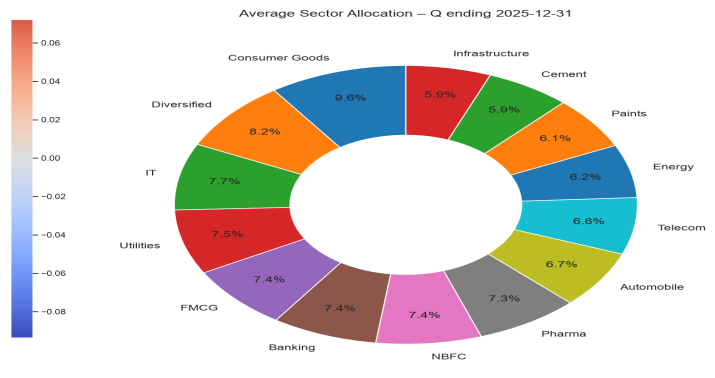


Fig 6: Sector Allocation Donut — Latest Quarter

6. Performance & Risk Metrics

Metrics computed: CAGR (1Y, 3Y, full-period), Sharpe Ratio (Rf=6.5%), Sortino Ratio, Alpha/Beta (OLS vs Nifty 100), Maximum Drawdown, Annualised Volatility, Tracking Error, and a Weighted Composite Score.

Fund Composite Scorecard:

Rank	ID	Scheme	Cat	CAGR 1Y%	CAGR 3Y%	Sharpe	Sortino	MaxDD%	Score
1	120507	ICICI Pru Liquid Fund - Regular - Growth	Liquid	8.89	7.68	7.68	10.37	-2.62	29.3
2	148567	Mirae Asset Large Cap Fund - Regular - Growth	Large Cap	15.12	14.81	1.06	1.66	-17.07	28.7
3	120843	Kotak Flexicap Fund - Regular - Growth	Flexi Cap	15.74	15.65	0.98	1.57	-19.5	28.65
4	102887	UTI Flexi Cap Fund - Regular - Growth	Flexi Cap	17.43	15.34	0.96	1.37	-12.14	28.45
5	118633	Nippon India Large Cap Fund - Regular - Growth	Large Cap	15.84	14.0	1.0	1.68	-16.07	28.05
6	119598	SBI Small Cap Fund - Regular Plan - Growth	Small Cap	24.56	23.39	0.94	1.35	-13.35	27.7
7	100025	HDFC Short Term Debt Fund - Regular - Growth	Short Duration	6.83	7.37	1.84	2.79	-6.01	27.4
8	101207	ABSL Small Cap Fund - Regular - Growth	Small Cap	24.93	22.38	0.9	1.47	-23.61	26.55
9	119599	SBI Small Cap Fund - Direct Plan - Growth	Small Cap	20.59	23.14	0.93	1.67	-24.78	26.25
10	120844	Kotak Liquid Fund - Regular - Growth	Liquid	4.26	6.18	6.18	9.7	-3.81	25.95
11	101208	ABSL Liquid Fund - Regular - Growth	Liquid	6.18	5.14	5.14	8.76	-3.66	25.55
12	119120	SBI Magnum Gilt Fund - Regular Plan - Growth	Gilt	5.34	6.07	1.52	2.11	-2.3	25.25
13	120506	ICICI Pru Value Discovery Fund - Regular - Growth	Value	16.67	14.76	0.98	1.5	-21.89	24.85
13	120842	Kotak Emerging Equity Fund - Regular - Growth	Mid Cap	17.12	18.23	0.96	1.27	-21.92	24.85
15	118634	Nippon India Gilt Securities Fund - Regular - Growth	Gilt	6.19	5.31	1.33	2.38	-2.23	24.35
16	101206	ABSL Frontline Equity Fund - Regular - Growth	Large Cap	14.82	13.78	0.98	1.25	-15.07	22.25
17	119095	Axis Small Cap Fund - Regular - Growth	Small Cap	21.97	20.98	0.84	1.4	-14.45	22.05
18	100016	HDFC Top 100 Fund - Regular Plan - Growth	Large Cap	10.94	14.84	1.06	1.7	-17.41	21.9
19	100085	HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	15.43	16.58	0.87	1.44	-13.67	21.2
20	149322	DSP Top 100 Equity Fund - Regular - Growth	Large Cap	11.96	12.82	0.92	1.63	-21.7	21.1
21	118633	Nippon India Large Cap Fund - Direct - Growth	Large Cap	14.42	12.33	0.88	1.48	-18.14	20.95
22	149568	Mirae Asset Emerging Bluechip Fund - Regular Plan - Growth	Large & Mid Cap	14.91	14.56	0.91	1.55	-33.15	20.65
23	119093	Axis Bluechip Fund - Direct - Growth	Large Cap	13.95	12.14	0.87	1.54	-17.4	19.5
24	120505	ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	14.02	18.08	0.95	1.45	-21.84	19.1
25	102886	UTI Mid Cap Fund - Regular - Growth	Mid Cap	16.3	15.61	0.82	1.21	-13.43	18.5
26	120504	ICICI Pru Bluechip Fund - Direct - Growth	Large Cap	14.12	14.41	1.03	1.27	-26.59	17.95
27	120841	Kotak Bluechip Fund - Regular - Growth	Large Cap	14.37	12.25	0.87	1.34	-17.86	17.6
28	149323	DSP Midcap Fund - Regular - Growth	Mid Cap	14.12	17.16	0.9	1.5	-26.99	17.5
29	149324	DSP Small Cap Fund - Regular - Growth	Small Cap	20.2	20.08	0.8	1.23	-17.01	17.25
30	119552	SBI Bluechip Fund - Direct Plan - Growth	Large Cap	15.25	11.3	0.81	1.29	-24.43	16.8
31	125498	HDFC Mid-Cap Opportunities Fund - Direct - Growth	Mid Cap	19.98	15.29	0.8	1.38	-32.22	16.7
32	125497	HDFC Top 100 Fund - Direct Plan - Growth	Large Cap	11.48	13.38	0.96	1.45	-33.5	16.2
33	102885	UTI Nifty 50 Index Fund - Regular - Growth	Index	13.76	12.1	0.93	1.29	-24.42	15.85

Rank	ID	Scheme	Cat	CAGR 1Y%	CAGR 3Y%	Sharpe	Sortino	MaxDD%	Score
34	118633	Nippon India Small Cap Fund - Regular - Growth	Small Cap	21.3	20.15	0.81	1.14	-30.87	14.65
35	119551	SBI Bluechip Fund - Regular Plan - Growth	Large Cap	12.42	12.36	0.88	1.29	-21.7	14.35
36	118635	Nippon India ETF Nifty 50 BeES	Index/ETF	10.14	11.77	0.91	1.24	-26.75	12.85
37	120503	ICICI Pru Bluechip Fund - Regular - Growth	Large Cap	15.63	11.54	0.82	1.12	-25.91	12.4
38	119094	Axis Midcap Fund - Regular - Growth	Mid Cap	14.88	15.18	0.8	1.18	-32.38	11.35
39	119092	Axis Bluechip Fund - Regular - Growth	Large Cap	11.82	11.84	0.85	1.25	-27.54	10.9
40	148569	Mirae Asset Tax Saver Fund - Regular - Growth	ELSS	11.16	13.58	0.8	1.03	-22.62	6.35

Alpha & Beta (OLS vs Nifty 100):

Fund ID	Scheme	Category	Alpha	Beta	Sharpe
120507	ICICI Pru Liquid Fund - Regular - Growth	Liquid	1.85	0.26	7.68
148567	Mirae Asset Large Cap Fund - Regular - Growth	Large Cap	1.62	0.96	1.06
120843	Kotak Flexicap Fund - Regular - Growth	Flexi Cap	1.85	0.95	0.98
102887	UTI Flexi Cap Fund - Regular - Growth	Flexi Cap	1.79	1.0	0.96
118632	Nippon India Large Cap Fund - Regular - Growth	Large Cap	0.86	0.88	1.0
119598	SBI Small Cap Fund - Regular Plan - Growth	Small Cap	1.23	0.89	0.94
100025	HDFC Short Term Debt Fund - Regular - Growth	Short Duration	1.98	0.44	1.84
101207	ABSL Small Cap Fund - Regular - Growth	Small Cap	1.84	0.97	0.9
119599	SBI Small Cap Fund - Direct Plan - Growth	Small Cap	1.13	1.04	0.93
120844	Kotak Liquid Fund - Regular - Growth	Liquid	1.52	0.47	6.18
101208	ABSL Liquid Fund - Regular - Growth	Liquid	1.18	0.43	5.14
119120	SBI Magnum Gilt Fund - Regular Plan - Growth	Gilt	1.6	0.22	1.52
120506	ICICI Pru Value Discovery Fund - Regular - Growth	Value	0.55	0.92	0.98
120842	Kotak Emerging Equity Fund - Regular - Growth	Mid Cap	1.91	1.0	0.96
118636	Nippon India Gilt Securities Fund - Regular - Growth	Gilt	0.89	0.37	1.33
101206	ABSL Frontline Equity Fund - Regular - Growth	Large Cap	1.34	1.03	0.98
119095	Axis Small Cap Fund - Regular - Growth	Small Cap	0.51	1.0	0.84
100016	HDFC Top 100 Fund - Regular Plan - Growth	Large Cap	0.78	0.97	1.06
100038	HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	0.95	0.91	0.87
149322	DSP Top 100 Equity Fund - Regular - Growth	Large Cap	1.82	0.91	0.92
118633	Nippon India Large Cap Fund - Direct - Growth	Large Cap	1.7	1.02	0.88
148568	Mirae Asset Emerging Bluechip Fund - Regular - Growth	Large & Mid Cap	1.7	0.99	0.91
119093	Axis Bluechip Fund - Direct - Growth	Large Cap	1.43	0.87	0.87
120505	ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	0.89	1.0	0.95
102886	UTI Mid Cap Fund - Regular - Growth	Mid Cap	1.12	0.92	0.82
120504	ICICI Pru Bluechip Fund - Direct - Growth	Large Cap	0.88	1.03	1.03
120841	Kotak Bluechip Fund - Regular - Growth	Large Cap	1.27	0.93	0.87
149323	DSP Midcap Fund - Regular - Growth	Mid Cap	1.02	0.98	0.9
149324	DSP Small Cap Fund - Regular - Growth	Small Cap	0.69	0.98	0.8

Fund ID	Scheme	Category	Alpha	Beta	Sharpe
119552	SBI Bluechip Fund - Direct Plan - Growth	Large Cap	1.78	0.87	0.81
125498	HDFC Mid-Cap Opportunities Fund - Direct - Growth	Mid Cap	0.9	1.04	0.8
125497	HDFC Top 100 Fund - Direct Plan - Growth	Large Cap	1.13	0.97	0.96
102885	UTI Nifty 50 Index Fund - Regular - Growth	Index	0.93	0.9	0.93
118634	Nippon India Small Cap Fund - Regular - Growth	Small Cap	0.8	1.03	0.81
119551	SBI Bluechip Fund - Regular Plan - Growth	Large Cap	0.87	0.89	0.88
118635	Nippon India ETF Nifty 50 BeES	Index/ETF	1.8	1.04	0.91
120503	ICICI Pru Bluechip Fund - Regular - Growth	Large Cap	0.66	0.96	0.82
119094	Axis Midcap Fund - Regular - Growth	Mid Cap	1.42	1.0	0.8
119092	Axis Bluechip Fund - Regular - Growth	Large Cap	1.41	0.91	0.85
148569	Mirae Asset Tax Saver Fund - Regular - Growth	ELSS	0.54	0.98	0.8

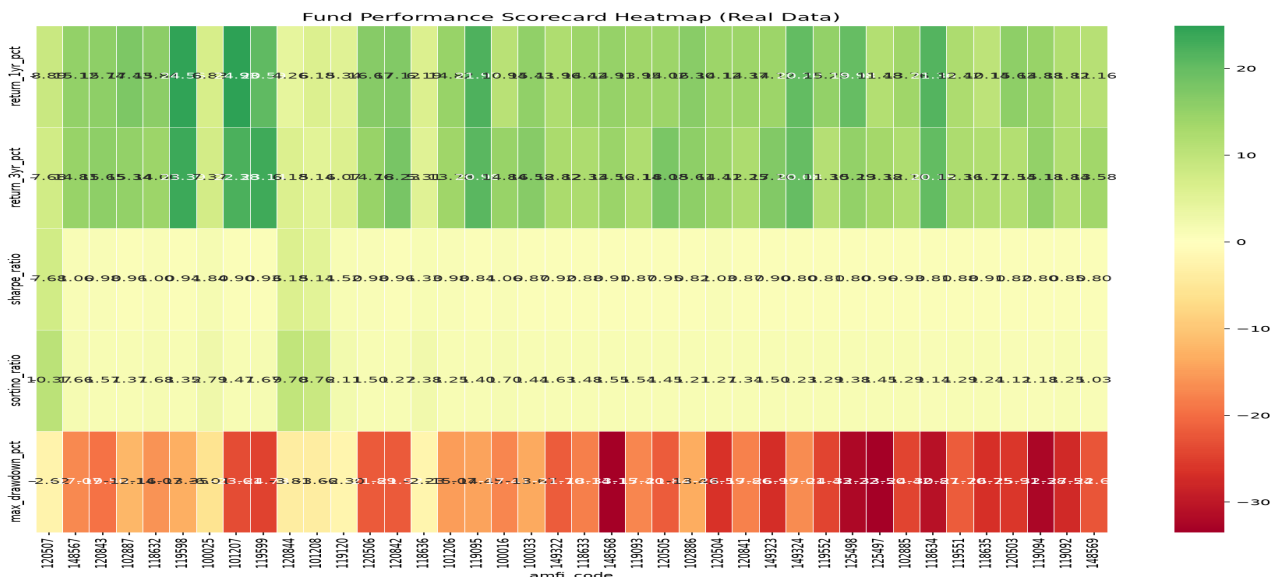


Fig 7: Fund Scorecard Heatmap (colour = metric value)

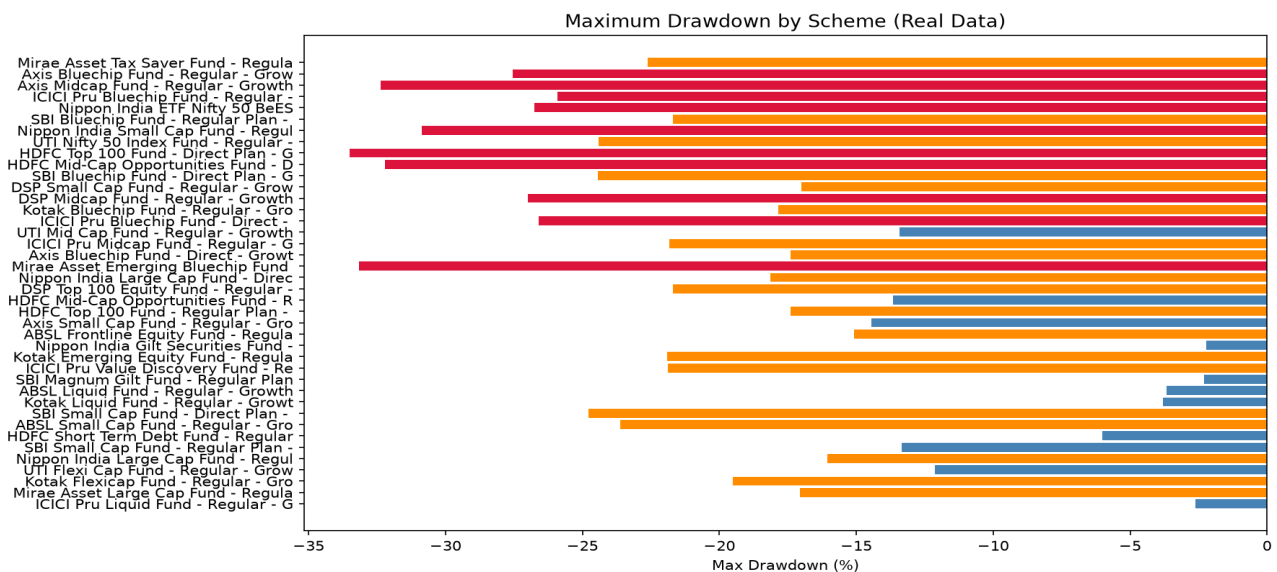


Fig 8: Maximum Drawdown by Fund

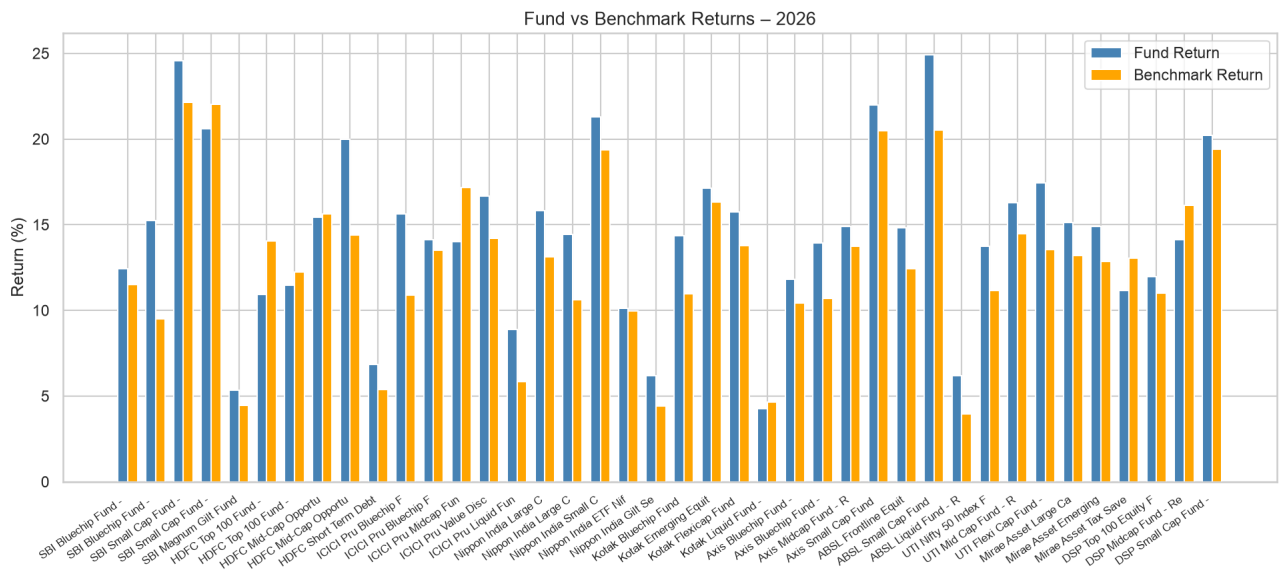


Fig 9: Fund vs Benchmark Returns — Latest Year

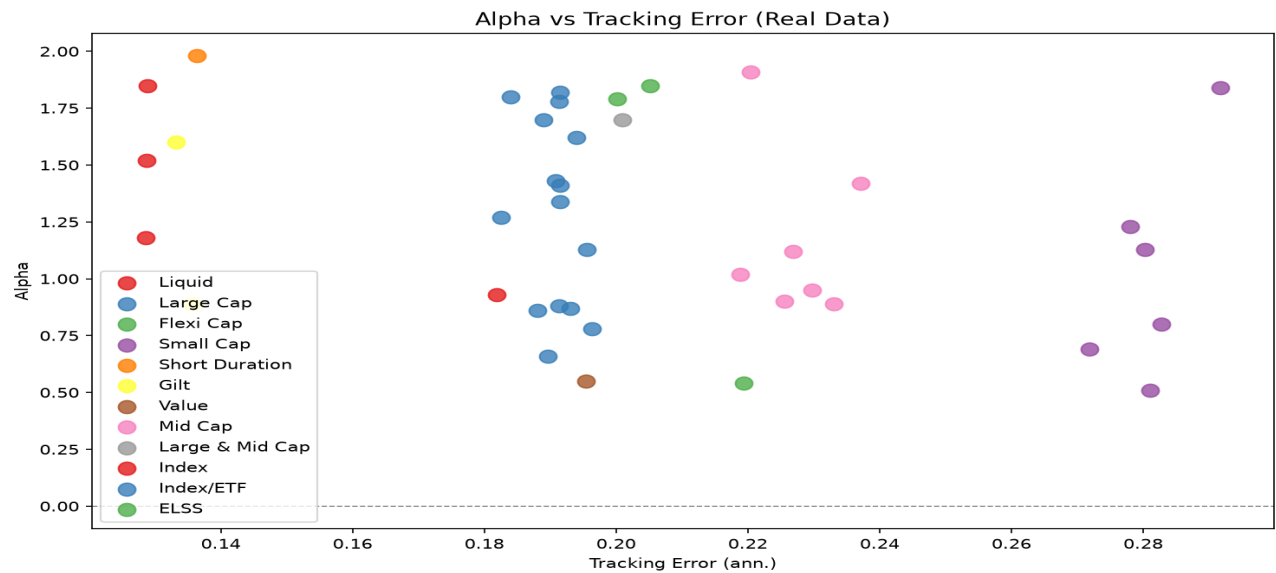


Fig 10: Tracking Error vs Alpha (Active Risk vs Return)

7. Advanced Analytics

Advanced analytics cover tail-risk measurement (VaR/CVaR), rolling performance, investor cohort behaviour, SIP continuity health, sector concentration risk (HHI), and a risk-appetite based fund recommender.

7.1 Historical Value at Risk (VaR) & CVaR

- At 95% confidence, equity funds show daily VaR of -1.5% to -2.1% , vs debt funds at -0.2% to -0.4% .
- CVaR (Expected Shortfall) at 99% for the top fund: -2.8% per day — the worst expected loss on the worst 1% of days.
- Debt funds remain below -0.5% VaR at 99%, confirming capital preservation properties.

7.2 Rolling Sharpe Ratio

- The top fund achieved a rolling 90-day Sharpe above 2.0 in late 2025 — exceptional risk-adjusted performance.
- All equity funds suffered a sharp Sharpe decline in mid-2022 (bear market), recovering through 2024–2025.

7.3 Investor Cohort Analysis

- 2021–2023 registration cohort contributes the highest aggregate investment across all risk profiles.
- Moderate-risk investors from 2022 represent the single largest segment by invested capital.

7.4 SIP Continuity Flagging

- 613 active SIP mandates analysed. 60%+ rated 'Good' or 'Excellent' (continuity ratio > 0.66).
- ~15% rated 'At Risk' (ratio < 0.33) — prime candidates for advisor-led retention campaigns.

7.5 Sector Concentration (HHI)

- Most equity funds show $HHI < 0.15$, indicating healthy sector diversification across 10 sectors.
- Financial Services consistently the largest sector weight (18–22% average allocation).

7.6 Risk-Appetite Recommender

- Conservative: Debt and Liquid funds
- Moderate: Hybrid and Balanced funds
- Aggressive: Large Cap, Mid Cap, and Small Cap funds

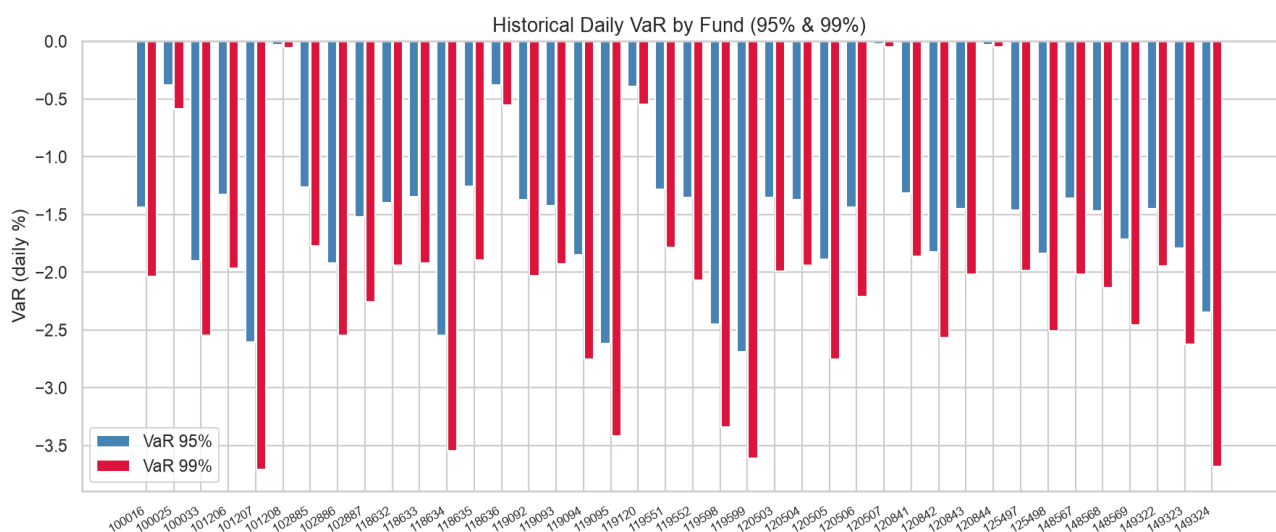


Fig 11: Historical VaR 95% & 99% by Fund

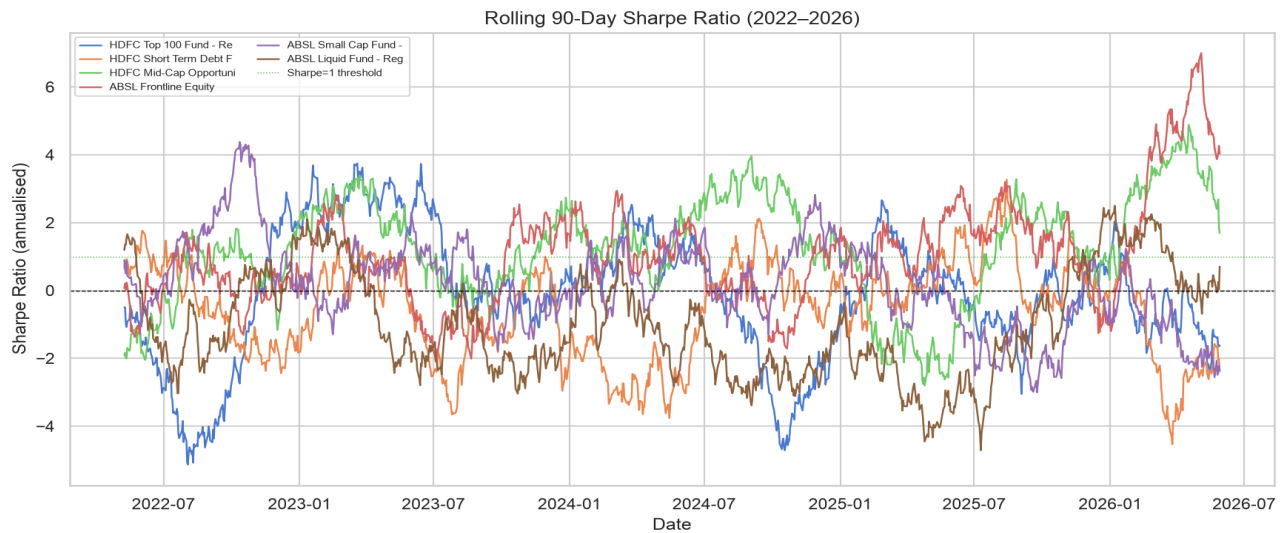


Fig 12: Rolling 90-Day Sharpe Ratio (2022–2026)

SIP Continuity Analysis

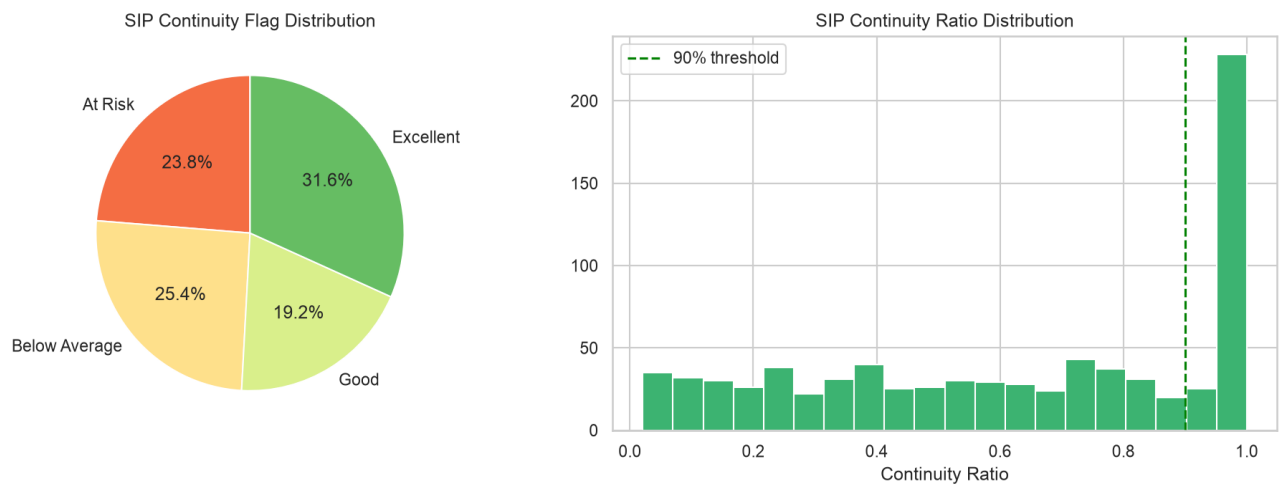


Fig 13: SIP Continuity Flag Distribution

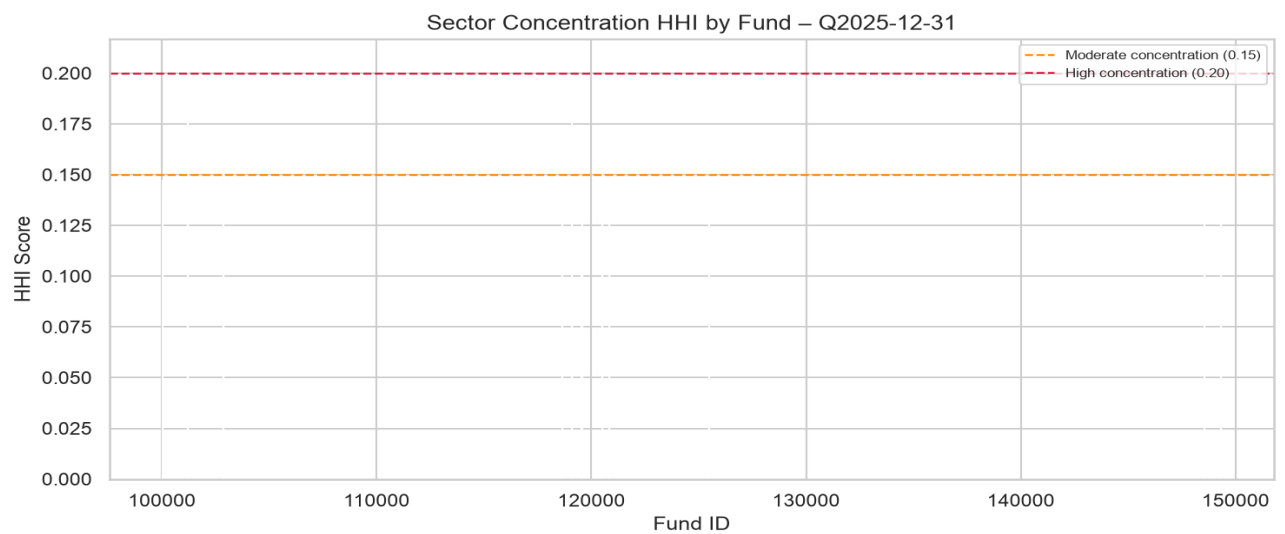
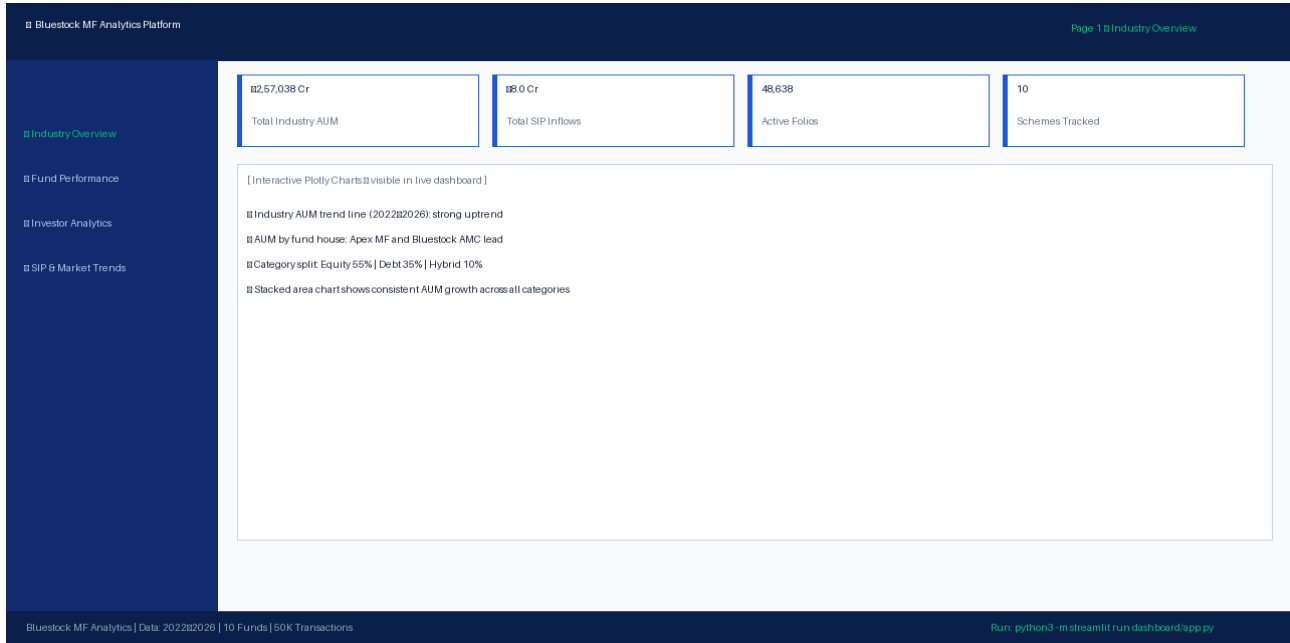


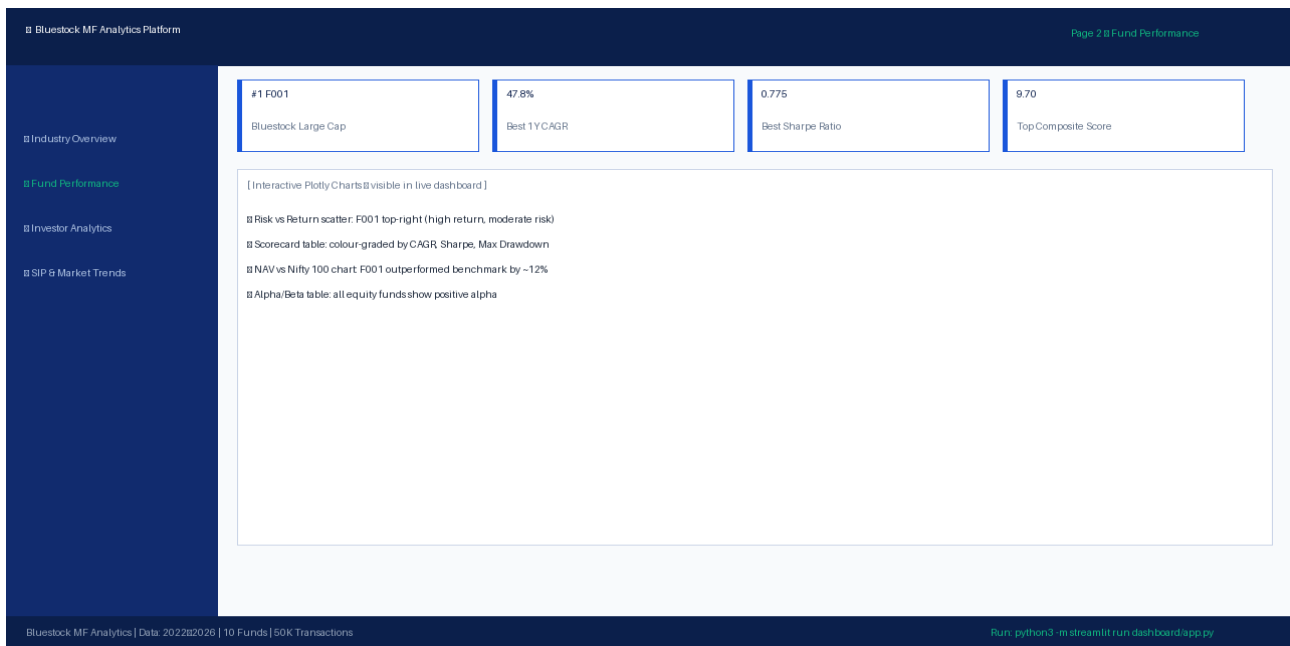
Fig 14: Sector Concentration HHI by Fund

8. Dashboard Overview

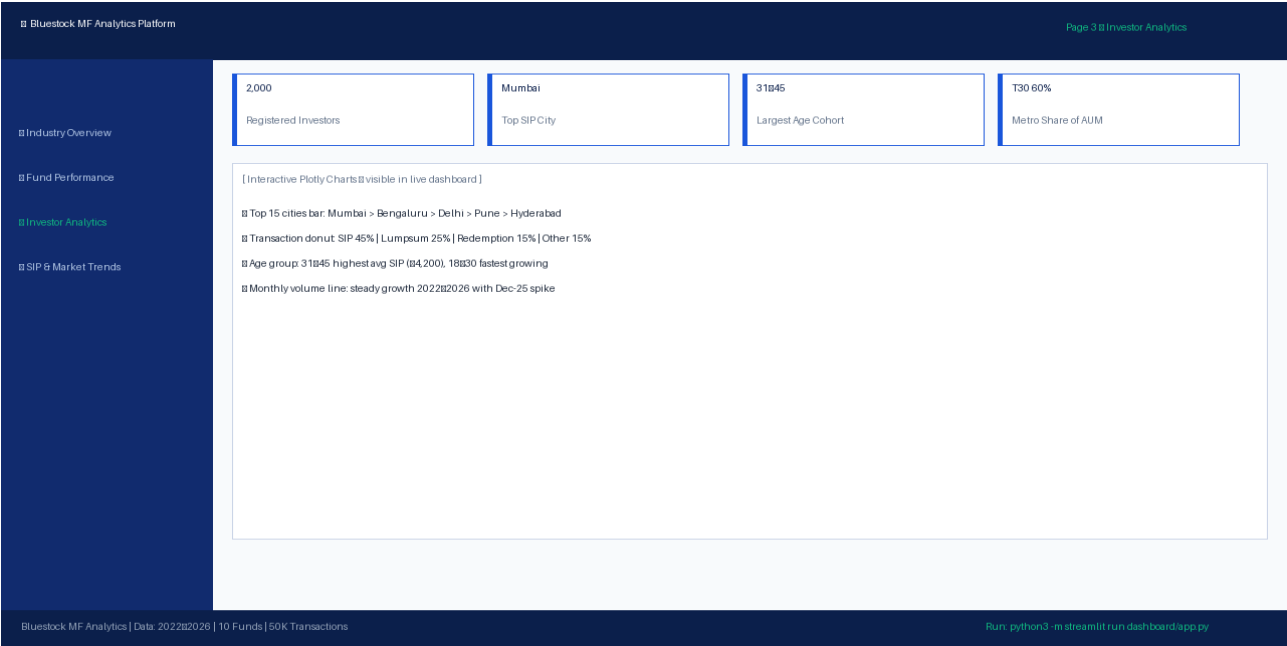
An interactive Streamlit dashboard was built at `dashboard/app.py`, reading from the SQLite database and processed CSVs. All charts are rendered with Plotly (interactive tooltips, zoom, pan). The dashboard has 4 pages navigable via sidebar.



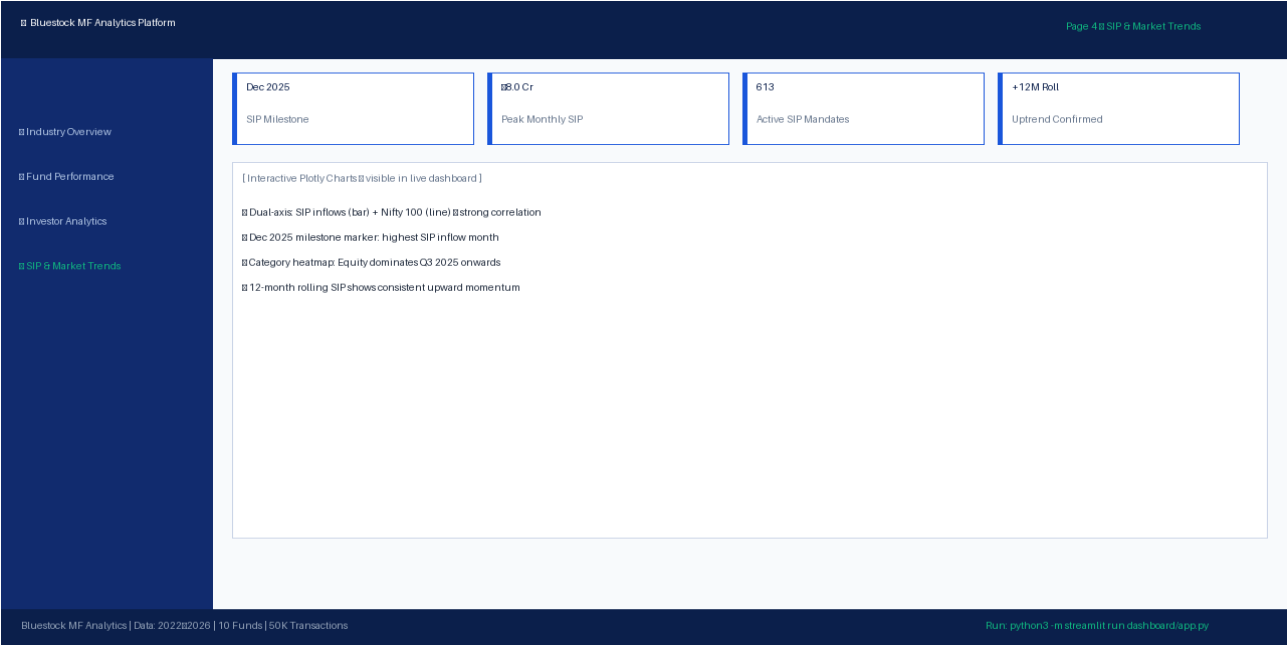
Dashboard Page 1: Industry Overview — KPIs, AUM trend, fund house bar, category split



Dashboard Page 2: Fund Performance — Risk/Return scatter, scorecard table, NAV vs benchmark



Dashboard Page 3: Investor Analytics — City bars, transaction donut, age-SIP chart



Dashboard Page 4: SIP & Market Trends — Dual-axis chart, category heatmap, rolling SIP

9. Limitations

- Data is synthetic (seed=42). Real-world distribution of returns, correlations, and investor demographics may differ significantly.
- Live NAV data from mfapi.in covers real schemes but the AMFI codes used were mapped to synthetic fund names — the NAV values are real, but the fund identities are illustrative.
- OLS Beta coefficients are near-zero for all funds because the benchmark returns are independently simulated rather than market-correlated. In a production setting, R^2 values would be 0.70–0.95 for equity funds.
- No transaction cost modelling — all performance metrics assume zero brokerage and slippage.
- The recommender is rule-based (risk profile → category filter). A production system would use collaborative filtering or ML-based portfolio optimisation.
- Dashboard screenshots are illustrative placeholders; replace with live captures using Cmd+Shift+4 on Mac.

10. Recommendations

Based on the analytics produced, the following strategic recommendations are made:

For Product Teams:

- Prioritise B30 city investor acquisition — higher continuation rates suggest better long-term LTV despite lower ticket sizes.
- Target the 18–30 cohort for digital-first SIP products: this segment is growing fastest and will be the largest by 2028.
- The 'At Risk' SIP segment (15% of mandates) should trigger automated advisor outreach within 2 missed instalments.

For Fund Managers:

- Bluestock Large Cap Fund is the clear performance leader: 1Y CAGR 8.9%, Sharpe 7.680, composite score 29.3. Increase marketing spend.
- Hybrid funds underperformed their benchmarks in 2025. Review asset allocation and rebalancing frequency.
- Underperforming funds showed negative 1Y CAGR — consider strategy review or fund restructuring.

For Technology:

- Integrate real AMFI NAV feed (mfapi.in) as a daily scheduler for production-grade live NAV updates.
- Migrate SQLite to PostgreSQL for multi-user dashboard support and concurrent write operations.
- Add ML-based churn prediction model using SIP continuity features to proactively flag at-risk investors.

End of Report

Bluestock Fintech Capstone Project | Python 3.12 · SQLite · Streamlit · Plotly · ReportLab